EXPERIMENT - 14

AIM:

To write a code using RAW sockets to implement packet sniffing.

CODE:

```
from scapy.all import sniff
from scapy.layers.inet import IP, TCP, UDP, ICMP

def packet_callback(packet):
    if IP in packet:
        ip_layer = packet[IP]
        protocol = ip_layer.proto
        src_ip = ip_layer.src
        dst_ip = ip_layer.dst

        protocol_name = ""
        if protocol == 1:
            protocol_name = "ICMP"
        elif protocol == 6:
            protocol_name = "TCP"
        elif protocol == 17:
            protocol_name = "UDP"
        else:
            protocol_name = "Unknown Protocol"

        print(f"Protocol: {protocol_name}")
        print(f"Source IP: {src_ip}")
        print(f"Destination IP: {dst_ip}")
        print("-" * 50)

sniff(iface = 'WiFi', prn = packet_callback,
      filter = "ip", store = 0)
```

OUTPUT:

Protocol : TCP
Source IP: 57.155.141.117
Destination IP: 192.168.1.6

Protocol : UDP
Source IP : 57.155.141.117
Destination IP: 192.168.1.1

Protocol : UDP
Source IP: 57.155.141.117
Destination IP: 192.168.1.1

Protocol: UDP
Source IP : 57.155.141.117
Destination IP : 192.168.1.1

RESULT:

Thus the program code using RAW sockets to implement packet sniffing was executed successfully.